AccessChain — SIH Master Pitch & Technical Architecture

Don't just find an accessible destination. Complete an accessible journey.

Executive Summary: *AccessChain is a world-class accessibility intelligence platform built for tourism and sports. It evaluates accessibility as an* ***end-to-end continuous journey property*** *rather than isolated place attributes, calculating explicit feasibility across transport, accommodation, venue entrance, restrooms, and seating for travelers with diverse accessibility profiles.*

1. Core Understanding of the Problem

Current accessibility platforms evaluate destinations independently. A hotel, metro line, airport, or stadium may individually claim to be accessible, but a person's **entire journey fails if even one connecting link is inaccessible** (e.g. non-wheelchair vehicle, broken lift without ramp alternative). Travelers face fragmented, static, and unreliable information across multiple apps, leading to travel anxiety and unexpected stranded situations.

2. What is the Solution?

AccessChain models origin, transit steps, transfers, venue gates, internal corridors, elevators, seating, and restrooms as connected **Nodes** and **Edges** in an **Accessibility Continuity Graph**. It answers: *'Can THIS SPECIFIC PERSON with THEIR EXPLICIT ACCESSIBILITY REQUIREMENTS complete THIS ENTIRE JOURNEY?'* If a link breaks or suffers a live outage, AccessChain automatically detects single points of failure (SPOFs) and calculates smart alternatives ('*Find A Way*').

3. What is Uniquely Present?

- **Accessibility Continuity Graph:** Node-edge spatial graph validating edge-by-edge constraints rather than simple place pin dropping.
- **Decoupled Three Core Metrics:** Decouples *FEASIBILITY* (FEASIBLE / NOT FEASIBLE), *ACCESSIBILITY QUALITY (0-100)*, and *JOURNEY RESILIENCE (0-100)*.
- **Tri-State Knowledge Engine:** Strictly enforces *UNKNOWN != FALSE* to prevent missing data from falsely assuming accessibility or inaccessibility.
- **Live Temporal Re-Evaluation:** Real-time graph updates (e.g., Gate 3 lift outage) trigger automatic re-evaluation via Gate 5 ramp.
- **SIH Interactive Demo Mode (/demo):** Deterministic 7-state demo machine simulating failure detection, 1-click shuttle repair, live outage, and rerouting.

4. What is Different from Other Platforms?

Feature / Aspect	Traditional Maps (Google Maps, TripAdvisor)	AccessChain (SIH Edition)
Evaluation Unit	Point of Interest (POI) / Isolated Pin	Continuous End-to-End Journey Graph
Routing Logic	Shortest time / distance	Safest Feasible Accessible Journey
Failure Detection	None (Discovered on arrival)	Proactive SPOF & Failure Diagnostics
Missing Data	Assumes accessible or ignores	Enforces UNKNOWN != FALSE & offers alternatives
Temporal Status	Static text labels	Real-time temporal status with outage rerouting

5. Benchmark: Ours vs Competitors

Benchmark Criteria	Google Maps	Wheelmap	AccessChain
End-to-End Continuity	No	No	100% Graph-Based
Multi-Constraint Matching	Partial	No	Full (Slope, Lift, Transit, Restroom)
Live Outage Rerouting	No	No	Real-Time Automated Reroute
SPOF & Resilience Metric	No	No	SPOF Counter & Resilience Score
Explainable Routing	No	No	Human-Readable Diagnostic Cards

6. Business Path & Revenue Model

- **B2C (Freemium):** Free basic journey discovery; Premium subscriptions for para-athletes & frequent travelers.
- **B2B (SaaS for Venues & Hotels):** Annual subscription for stadium operators, hotels, and event organizers to monitor live infrastructure and maintain accessibility compliance.
- **B2G (Government & Cities):** City-wide accessibility gap heatmaps & urban planning intelligence licenses.
- **API Infrastructure Layer:** Data licensing API for booking platforms (MakeMyTrip, Booking.com), airlines, railways, and map providers.

7. Tools Used in Development

- **IDE & Workspace:** Antigravity IDE (Gemini 3.6 Flash / Pro multi-agent execution engine).
- **Runtime & Package Manager:** Node.js v24, npm 11, TypeScript Compiler (*tsc*).
- **Testing & Linters:** Vitest unit testing framework, PostCSS, Tailwind CSS JIT compiler.

8. Tech, Frameworks, and Algorithms Used for SIH

- **Frontend & UI:** Next.js 14+ (App Router, React 19, TypeScript), Tailwind CSS, Lucide Icons, Framer Motion, Canvas Confetti.
- **State Management:** Zustand (Global profile manager, graph engine state, SIH demo state machine).
- **Database Schema:** Prisma ORM with PostgreSQL + PostGIS spatial data baseline.
- **Routing Algorithm (Dijkstra / A* Multi-Constraint):** Evaluates candidate graph edges, filtering infeasible links before performing multi-objective optimization for time, cost, comfort, and resilience.
- **Scoring Algorithm:** Weighted average formula across 8 sub-scores (Mobility, Transport, Accommodation, Venue, Restroom, Navigation, Sensory, Emergency).
- **Resilience Algorithm:** Single Point of Failure (SPOF) counter & graph redundancy metric.
- **SIH Demo State Machine:** 7-state deterministic runner (*INITIAL* ➔ *FAILED* ➔ *FIXING* ➔ *FIXED* ➔ *OUTAGE* ➔ *REROUTING* ➔ *RESTORED*).